Medical Engagement Workbench

A focused workflow for repeat-meeting preparation, scientific evidence review and follow-up.

3 bounded agents | Source-linked drafts | Human review

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Local working application | Synthetic CRM | Live public registry search

Explore the walkthrough | Open local app



Continuity across scientific conversations

Before the next meeting, the MSL needs prior context, relevant evidence and the unresolved request in one place.

What happened?

- Prior discussions and observations
- Outstanding tasks and medical requests
- An exact, dated KOL record

What supports it?

- Literature and registered studies
- Explicit source selection
- Original identifiers and text

What happens next?

- A structured draft note
- Open questions and requests
- MSL review and follow-up

> Value hypothesis: less reconstruction effort and fewer missed commitments. Time savings and usefulness have not yet been measured.

A complete local workflow, with clear boundaries

Three workspaces connect the CRM snapshot to public evidence and reviewable drafts.

3	9	9	6
KOL profiles	interactions	tasks	medical requests

01 | KOL context

Profile, interaction history, follow-up tasks and medical requests.

02 | Evidence search

PubMed and ClinicalTrials.gov filters, summaries and source selection.

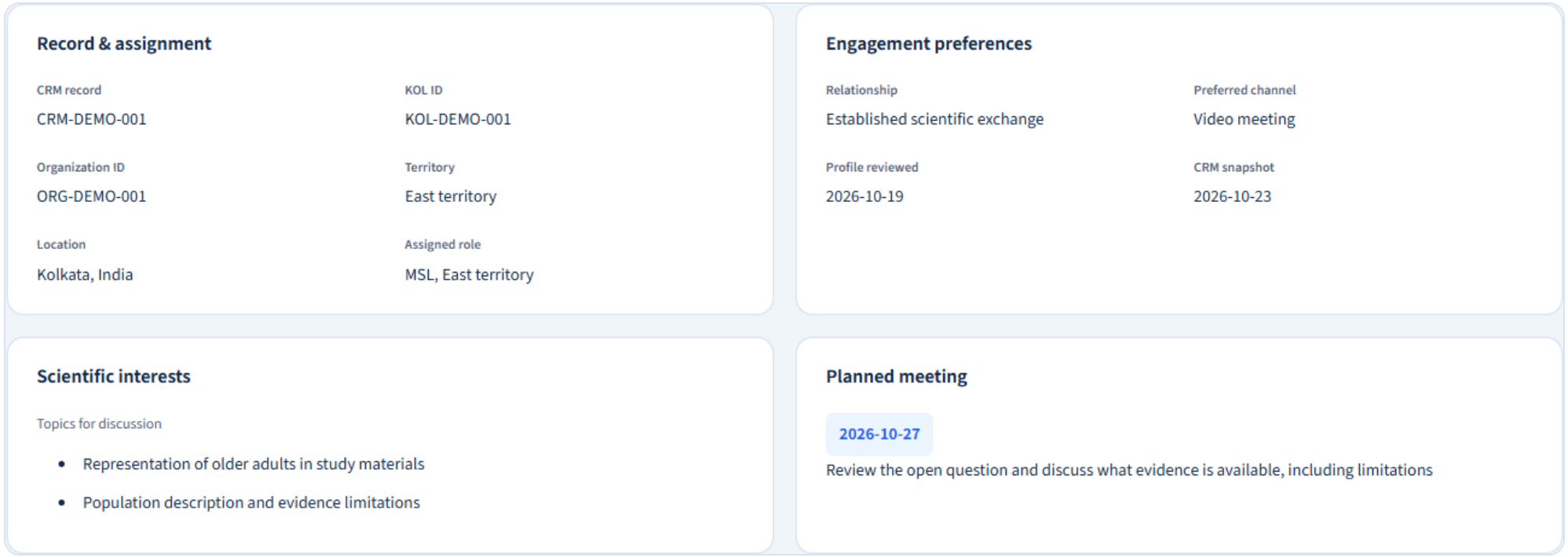
03 | Drafts & follow-up

Priorities, briefing, note drafts and downloadable preparation records.

CRM snapshot: 23 October 2026. Read-only fixture data; no live CRM connection or task write-back.

Know the KOL before reviewing the evidence

A profile organized around assignment, engagement preferences, scientific interests and the planned meeting.

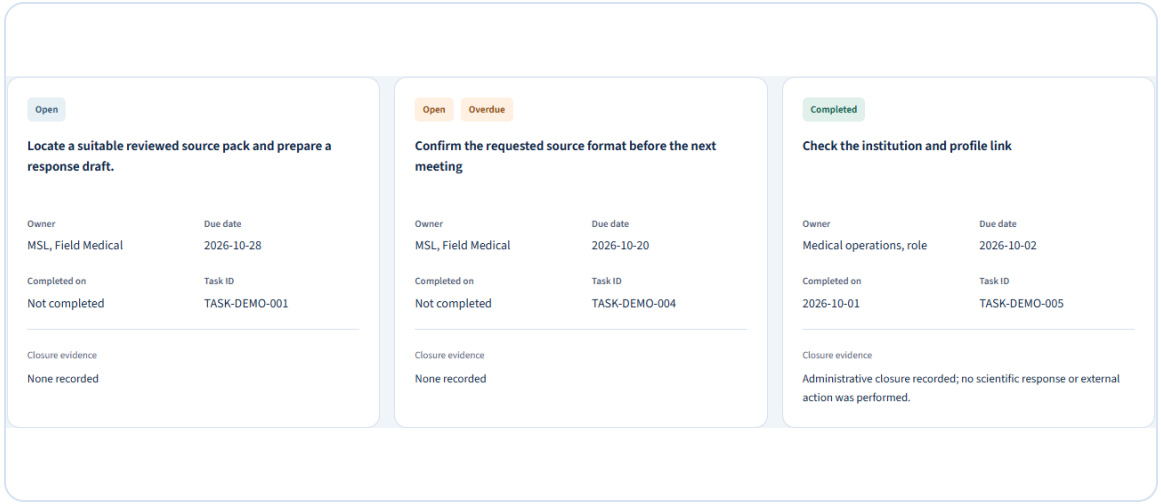


CRM profile: side-by-side record, engagement and meeting cards

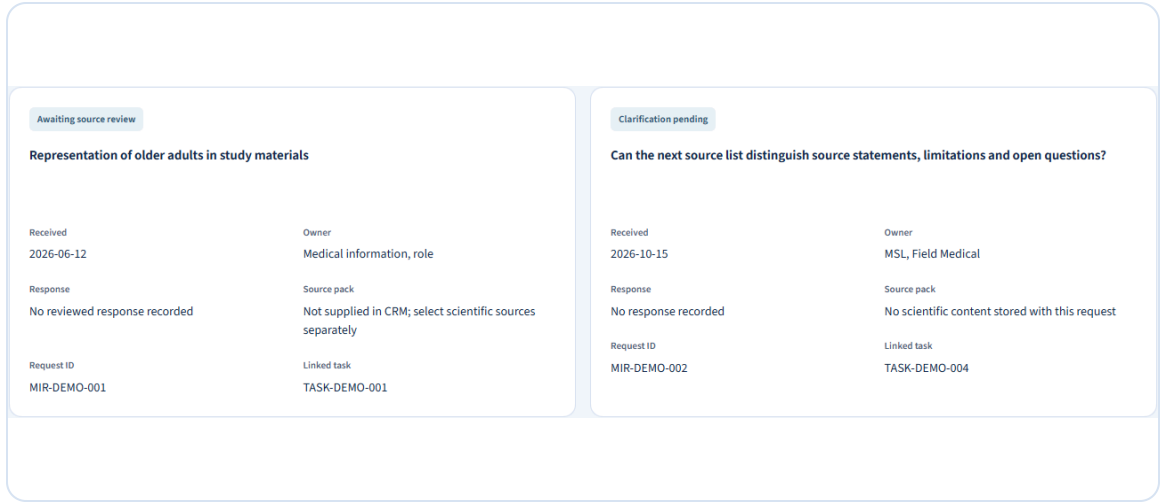
Exact-ID retrieval | Separate KOL records | Workspace clears on KOL change

Keep the next action and the open question visible

Read-only cards make status, ownership, dates and medical-information requests easier to scan.



Follow-up tasks: status, owner, due date and closure evidence

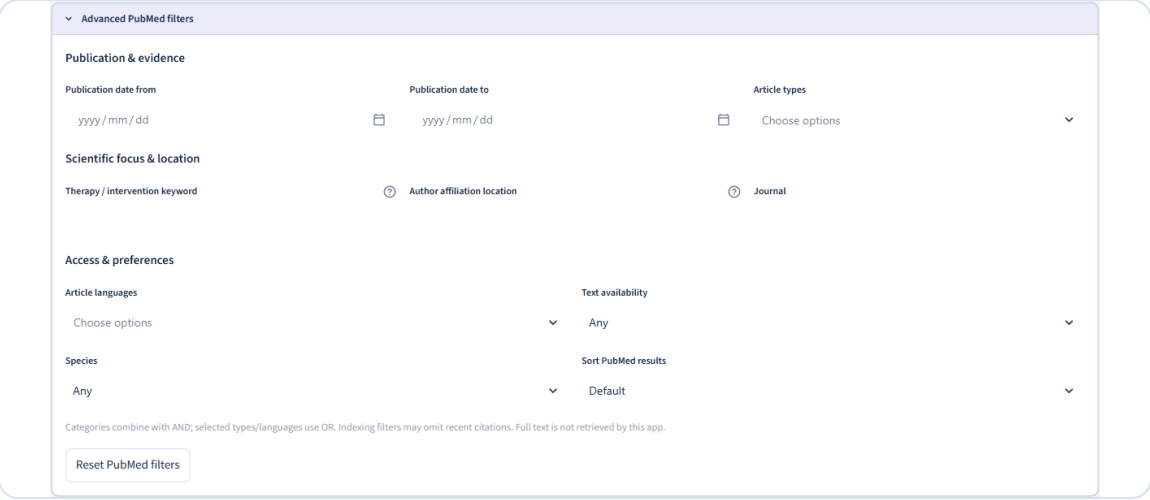


Medical requests: question, response status, source pack and linked task

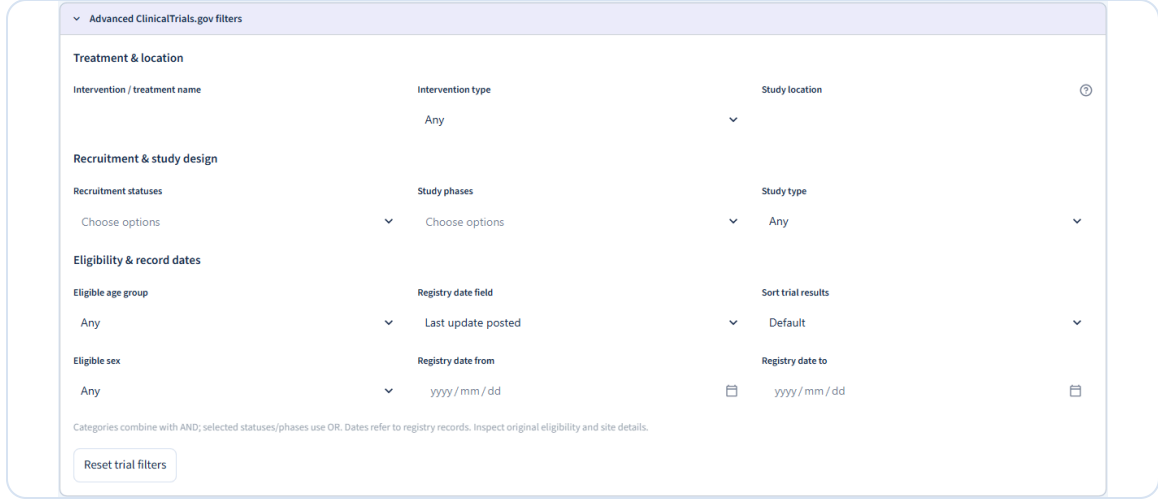
> Overdue uses the CRM snapshot date. Newly drafted requests do not assign an owner, set a deadline or change CRM status.

Filter the right source, using the right meaning

Filters are sent to the public API before the returned-record limit is applied.



PubMed publication, evidence, focus and access filters



ClinicalTrials.gov treatment, recruitment, design, eligibility and date filters

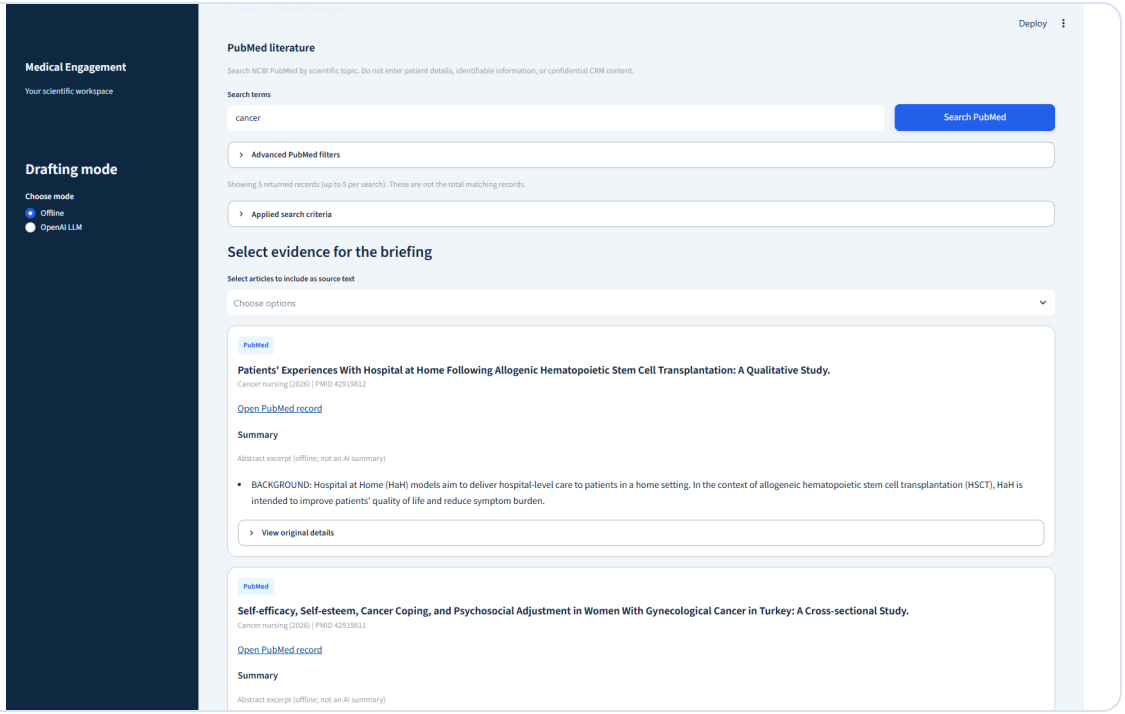
PubMed: publication dates, article types, languages, availability, species, therapy keywords, affiliation and journal.

Trials: treatment, location, recruitment, phase, design, eligibility and registry dates.

> Author affiliation is not study location. Trial registration/update dates are not publication dates. Up to five returned records per search is not an exhaustive export.

A short overview, with the original evidence available

Summaries support scanning. The briefing uses the original explicitly selected records.



Actual public-search snapshot: Offline excerpt, not an AI-generated summary

What the user reviews

- PMID or NCT ID and original link
- Offline excerpt / overview, or an explicit OpenAI summary request
- Original abstract or registry details under an expander
- Explicit source selection before preparation

> PubMed retrieval is metadata/abstract based. Planned trial outcomes are not represented as study results.

One responsibility per agent

Exactly three application agents, coordinated by LangGraph.

01

KOL Context Agent

INPUT

Selected KOL ID and CRM history.

OUTPUT

Exact context; OpenAI mode adds meeting priorities and unresolved requests.

CRM is context, not scientific evidence.

02

Scientific Source Agent

INPUT

Explicit searches and selected scientific sources.

OUTPUT

Retrieved records, summaries and a source-linked briefing with questions and limitations.

Missing sources trigger abstention.

03

Follow-up Agent

INPUT

MSL-entered meeting notes.

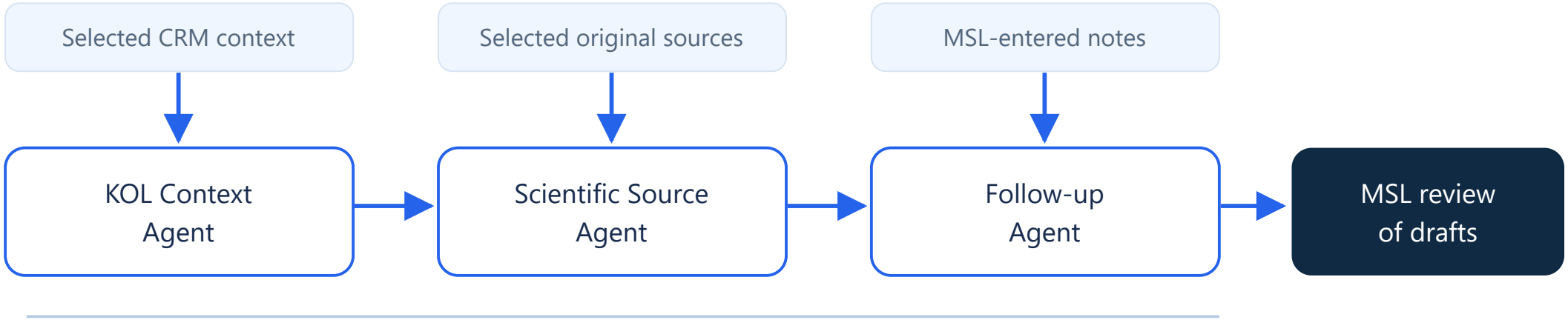
OUTPUT

Preserved notes plus draft observations, requests and open questions.

New owners and dates remain unassigned.

A deterministic flow around bounded model calls

The user chooses the KOL, initiates searches, selects evidence and explicitly requests preparation.



LangGraph coordinates the three preparation steps in order

Search time

Read-only public registry retrieval. In OpenAI mode, one public-record summary batch per nonempty search.

Preparation time

Offline or OpenAI drafting. Maximum three model requests; absent sources or notes can skip individual requests.

Designed to make unsupported output visible

Controls help the software fail clearly; they do not establish medical correctness.

Before and during generation

- Exact-ID lookup and input-size limits
- Strict structured outputs and schema validation
- Selected-source IDs and exact quote checks
- Bounded output, timeout and sanitized errors

At the review boundary

- Abstain without scientific evidence
- Clear stale drafts after failure or changed criteria
- Preserve original notes and scientific sources
- Leave approval and external action with people

> Matching a quote does not prove correct interpretation. No live CRM writes, external messages or safety submissions are implemented.

Software behavior tested, with separate evidence types

Actual saved execution artifacts | 24 October 2026

126
automated test cases passed

16
fixed harness cases passed

8
desktop/mobile checks passed

Automated coverage

- Agent contracts and graph order
- KOL isolation and UI recovery
- API parsing, filters and source selection
- Schema, citation and quote rejection

Observed browser checks

- Task and request card alignment
- Live PubMed and trial retrieval
- Grouped filters and visible summaries
- Mobile layouts without page overflow

> Harness cases overlap with unit-test behavior. These totals are not independent samples of model quality or a clinical-accuracy score.

A repeatable experiment around the actual agents

Local fake transport exercises production orchestration and parsing without API keys or network access.

01

Fixed cases

Synthetic inputs and explicit expected outcomes

02

Scripted transport

Structured local responses and error cases

03

Production graph

The same three agents and schema validation

04

Named checks

Identity, role inputs, evidence and call budget

05

Saved reports

JSON, readable results, versions and input hashes

Deliberate failure cases

Unknown identity, oversized input, missing evidence, unselected citations, invented quotes, refusals and malformed output.

Test the grader, too

Corrupt identity, notes, sources, credentials, role inputs, call order and other-KOL context; the grader must reject them.

Behavior contracts, implementation and evidence are separated

The repository is also the Obsidian vault, keeping the supporting documentation together.

Application structure

```
prototype/
  app.py                  Workspaces + session state
  agents.py               Three bounded roles
  workflow.py             LangGraph orchestration
  llm.py                  Model client + validation
  pubmed.py / clinical_trials.py
  search_filters.py       Query construction
  ui.py / crm_views.py    Presentation
```

Specification & QA structure

```
prototype/
  specs/                  Behavior contracts
    requirement-test-matrix.md
    agent-harness.md
    evaluation-and-review.md
  tests/                  Executable assertions
  harness/                Fixed-case evaluation
  test_scenarios/         Use cases + QA evidence
wiki/                     Project knowledge
```

One QA command: `./prototype/.venv/Scripts/python.exe -B prototype/harness/run_local_qa.py`

A reliable path through the working application

Start in Offline mode. Use OpenAI mode only for a separately configured and honestly labeled live run.

1. **Context | 1 minute**
   Select Dr Mira Sen. Show the profile, prior history, open task and medical request.
2. **Evidence | 2 minutes**
   Run public searches. Inspect a summary and original details; explicitly select a record.
3. **Drafts | 1 minute**
   Enter synthetic meeting notes and prepare. Explain Offline versus OpenAI behavior.
4. **Boundaries | 1 minute**
   Show abstention without selected evidence and workspace clearing on KOL change.
5. **Evidence | 1 minute**
   Show saved QA artifacts and identify remaining evaluator review.

Use the screenshot walkthrough | Open local app

The next milestone is reviewed evidence of usefulness

The local build is ready to demonstrate. Model quality and stakeholder acceptance need their own evidence.

Validate the outputs

- Record a local OpenAI walkthrough
- Check paraphrases against original sources
- Exercise missing/conflicting evidence and hostile source text

Complete evaluator review

- Name the reviewer and record decisions
- Score E1-E8 with the prepared rubric
- Measure a baseline before claiming time savings

Decide the next scope

- Agree approval and task-tracking needs
- Assess production identity and durable records
- Confirm a permitted host with the team

> Hosting remains on hold. No external public app URL has been created. The shareable HTML walkthrough works independently of hosting.

A presentation the team can inspect and share

Use the HTML file for interactive presenting and the PDF for circulation.

How to share

- Send index.html: screenshots are embedded
- Explore the walkthrough without a key or network calls
- Use presentation.pdf for static review
- The localhost app link works only on the presenter's computer

Supporting artifacts

- Automated QA report
- Agent-harness results
- Browser verification
- Requirement-to-test matrix
- Evaluator protocol

Explore the walkthrough

Print / save PDF

This is a dated evidence snapshot. Rerun QA and rebuild after later application changes.